

KARNATAKA 2ND PUC BOARD PREPARATION

Chapter 9: Biotechnology - Principles and Processes Official Blueprint Updated Study Guide

BAIO LEARNING SYSTEM | Narendra Naslai - Founder & CEO

Subject: Biology (36) | Weightage: 6 Marks

1. OFFICIAL KARNATAKA 2ND PUC CHAPTERWISE ALLOTMENT

PARAMETER	OFFICIAL ALLOTMENT DATA (CHAPTER 9)
Chapter Name	Chapter 9 - Biotechnology: Principles and Processes
Teaching Hours Allotted	6 Hours (Periods)
Official Marks Allotted	6 Marks
Page Numbers in Official Notes	Pages 163 – 176
Overall Paper Format	3 Hours Max Marks: 70 Total Allotted Weightage: 105 Marks (including choices)

Expected 6-Mark Question Distribution Configurations

- **Pattern A:** 1 Five-Mark Question (Part D) + 1 One-Mark MCQ/VSA (Part A). **Most Common**
- **Pattern B:** 1 Three-Mark Question (Part C) + 1 Three-Mark Question (Part C).
- **Pattern C:** 1 Three-Mark Question (Part C) + 1 Two-Mark Question (Part B) + 1 One-Mark Question (Part A).

2. TOP HIGH-YIELD TOPICS TARGETED FOR 6 MARKS ALLOTMENT

TOPIC 1 [5-Mark Essential]: Polymerase Chain Reaction (PCR) - Steps & Components

Frequently asked directly as a 5-Mark question in Part D: "What is PCR? Describe the three major steps in PCR amplification with a neat diagram."

- **Definition:** In vitro enzymatic synthesis of multiple copies of a specific gene/DNA segment using primers and thermostable DNA polymerase.
- **Step 1: Denaturation (94–96°C):** High temperature melts double-stranded DNA into two single template strands by breaking hydrogen bonds.
- **Step 2: Annealing (50–65°C):** Two sets of chemically synthesized oligonucleotide primers hybridize/anneal to complementary sequences at 3' ends of template strands.
- **Step 3: Extension (72°C):** Thermostable *Taq* DNA polymerase (isolated from *Thermus aquaticus*) synthesizes new DNA strands using dNTPs in 5' → 3' direction.
- **Result:** ~30 cycles amplify the target gene segment approximately 1 billion times (10^9 copies).

TOPIC 2: Features of Cloning Vector pBR322 & Selectable Markers (3 or 5 Marks)

Core Question: "Draw a neat labeled diagram of *E. coli* cloning vector pBR322 and explain its essential features."

- **Essential Components:**
 1. **Origin of Replication (ori):** Sequence where replication starts; controls copy number of linked DNA.
 2. **Selectable Markers:** Genes conferring antibiotic resistance (amp^R and tet^R) to identify/select transformants and eliminate non-transformants.
 3. **Restriction/Cloning Sites:** Unique recognition sites like *Bam*HI, *Sal*I, *Eco*RI, *Cl*aI, *Hind*III, *Pvu*I, *Pst*I.
 4. **rop:** Codes for proteins involved in plasmid replication.

- **Insertional Inactivation (Alternative Marker):** Recombinant DNA inserted into coding sequence of β -galactosidase inactivates it. Recombinants form white colonies, while non-recombinants produce blue colonies in presence of chromogenic substrate.

TOPIC 3: Isolation of DNA & Agarose Gel Electrophoresis (3 or 5 Marks)

- **Cell Lysis Enzymes:** Lysozyme (bacteria), Cellulase (plant cells), Chitinase (fungi). RNase removes RNA; Protease removes proteins.
- **Precipitation:** Pure DNA precipitates out as fine threads upon addition of **chilled ethanol** (retrieved by *spooling*).
- **Gel Electrophoresis:** Negatively charged DNA fragments migrate toward anode (+) through agarose matrix based on size (sieving effect; smaller fragments move faster).
- **Visualization & Elution:** Stained with **Ethidium Bromide (EtBr)** and exposed to UV light to see bright orange bands. Cutting out and extracting DNA band from gel is **Elution**.

TOPIC 4: Bioreactors & Downstream Processing (2, 3, or 5 Marks)

- **Simple vs. Sparged Stirred-Tank Bioreactor:** Vessels for processing large volumes (100–1000 L) of culture. Features agitator system, oxygen delivery, foam breaker, temperature/pH control, and sampling ports.
- **Downstream Processing:** Collective term for separation and purification processes required before marketing a product, followed by formulation with preservatives and strict quality control testing.
- **Gene Transfer Methods:** Micro-injection (animal cells), Biolistics / Gene Gun (plant cells with gold/tungsten particles), Heat Shock (42°C with CaCl_2), Disarmed Pathogens (*Agro. tumefaciens* / Retroviruses).

3. MODEL QUESTIONS MATCHED TO 6-MARK BLUEPRINT COMBINATION

Q1. Which enzyme is used to precipitate purified DNA from cellular lysate suspension?

1 Mark (Part A MCQ)

Model Answer: Addition of chilled ethanol (precipitates DNA which is collected by spooling). [1 Mark]

Q2. Name the source organism of thermostable Taq DNA polymerase enzyme used in PCR.

1 Mark (Part A VSA)

Model Answer: *Thermus aquaticus* (a thermophilic bacterium). [1 Mark]

Q3. What is insertional inactivation? Mention its significance in gene cloning.

2 Marks (Part B)

Model Answer: Inactivation of a gene (e.g., β -galactosidase or antibiotic resistance gene) due to insertion of recombinant foreign DNA within its coding sequence. It helps in easy differentiation between recombinants (white colonies) and non-recombinants (blue colonies). [2 Marks]

Q4. Describe the steps involved in Agarose Gel Electrophoresis for separation and visualization of DNA.

3 Marks (Part C)

Model Answer:

1. **Separation:** Negatively charged DNA fragments move towards anode (+) through agarose gel matrix under electric field, separating according to size. [1 Mark]
2. **Visualization:** DNA fragments are stained with Ethidium Bromide (EtBr) and exposed to UV radiation to view bright orange bands. [1 Mark]
3. **Elution:** The separated DNA bands are cut out from agarose gel and extracted from gel piece for cloning. [1 Mark]

Q5. Explain Polymerase Chain Reaction (PCR) with reference to its 3 steps and mention the importance of Taq polymerase.

5 Marks (Part D)

Model Answer:

1. **Denaturation:** Double-stranded target DNA is heated to 94–96°C to separate into two single strands. [1 Mark]
2. **Annealing:** Two sets of oligonucleotide primers hybridize to complementary regions at 3' ends of template strands at 50–65°C. [1 Mark]
3. **Extension:** *Taq* DNA polymerase extends primers using nucleotides at 72°C to synthesize new complementary strands. [1 Mark]
4. **Taq Polymerase Role:** Isolated from *Thermus aquaticus*; remains stable/active during high-temperature denaturation, enabling automated repeated cycles (~1 billion copies). [1 Mark]
5. **Diagram / Summary:** Schematic representation showing Denaturation → Annealing → Extension leading to amplified gene copies. [1 Mark]

BLUEPRINT SCORING TIP FOR KARNATAKA 2ND PUC

Since Chapter 9 carries **6 Marks** with **6 Teaching Hours**, master the complete 5-mark answer for PCR or pBR322 vector diagram along with 1-mark objective questions on enzymes (*Taq* polymerase, Restriction enzymes, Ligase) and specialized techniques (*Elution*, *Spooling*, *Downstream processing*).